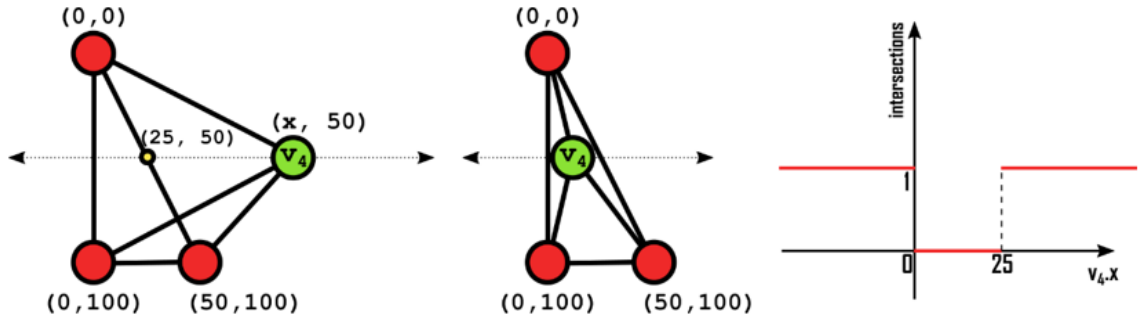


Figure 16.7 The cost function for "number of intersections" for a specific embedding of K_4. The position of 3 out of 4 vertices is locked, and for the last one, v_4 in the graphic, we can only change its horizontal position. Under these assumptions, the cost as a function of v_4's `x` position is a discontinuous function, equal to `0` when `0 < x < 25`, and equal to 1 when `x < 0` or `x > 25`. Notice how at `x==0` and `x==25` we have two discontinuity points, and in particular we assume the cost is infinite at those points, because the vertex is exactly lying on one of the edges between the other vertices.

The cost function we have chosen for our graph embedding problem is simply the number of edge intersections for a given embedding. Spoiler alert: this choice is problematic with some optimization algorithms, as we'll later see.

So, in figure 16.7 we can see the complete graph K_4, an embedding that has 8 degrees of freedom: let's agree to lock 7 of them, and only allow to change the horizontal position of vertex "v_4".

The cost function looks like a *step function*, with its value going abruptly from `1` to `0` when the vertex enters the internal face of the sub-graph induced by the other three vertices (the triangle marked by the red, unnamed vertices in figure 16.7).